CDE2310 G2 End User Documentations

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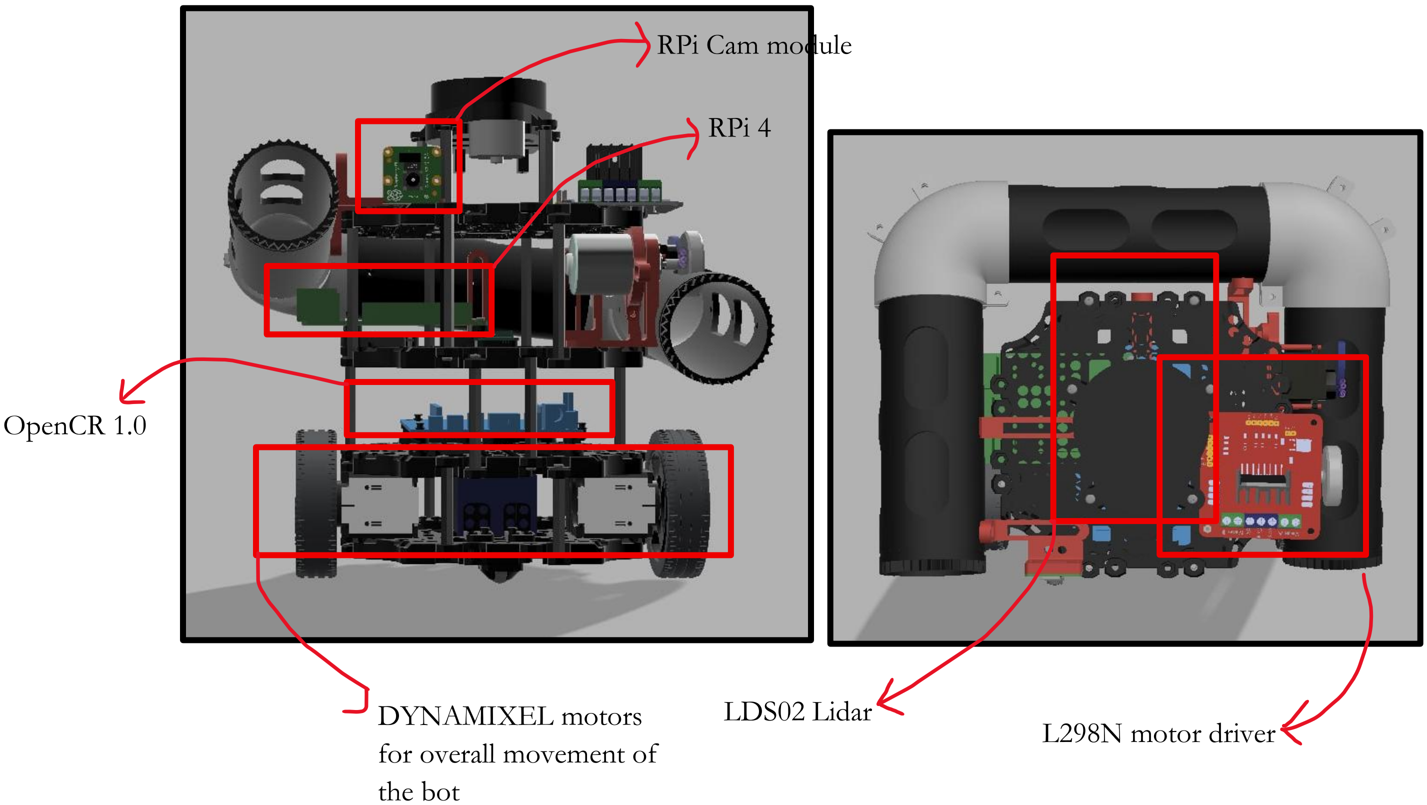
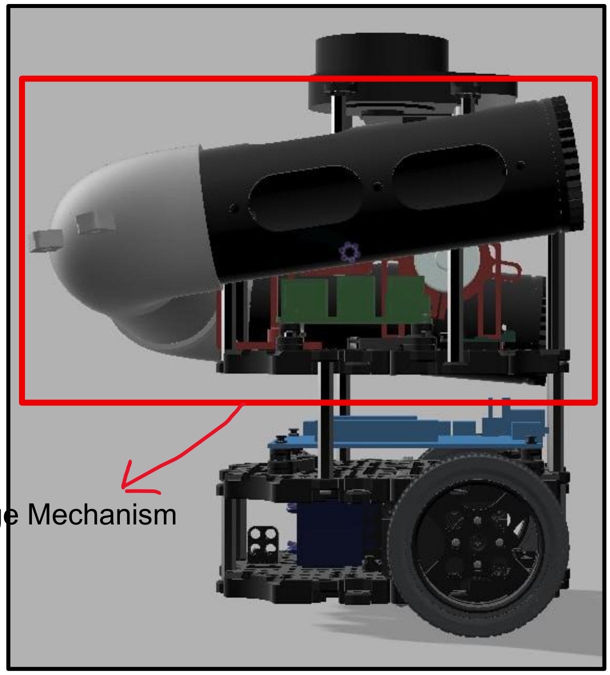
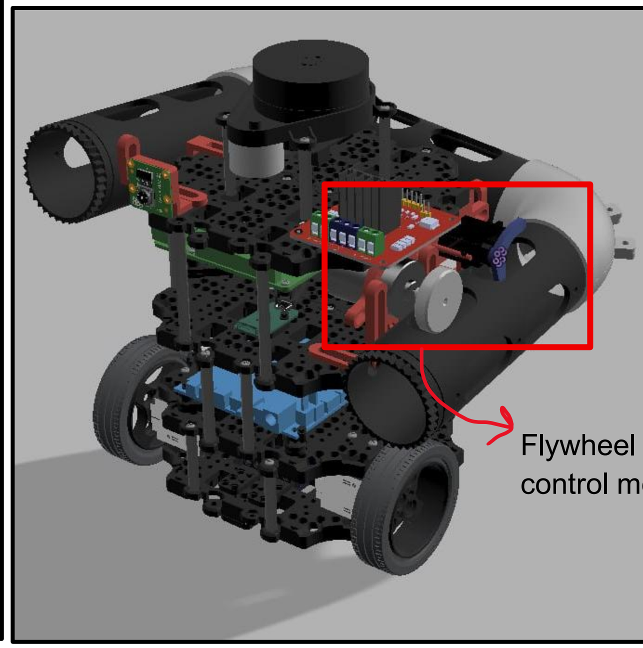
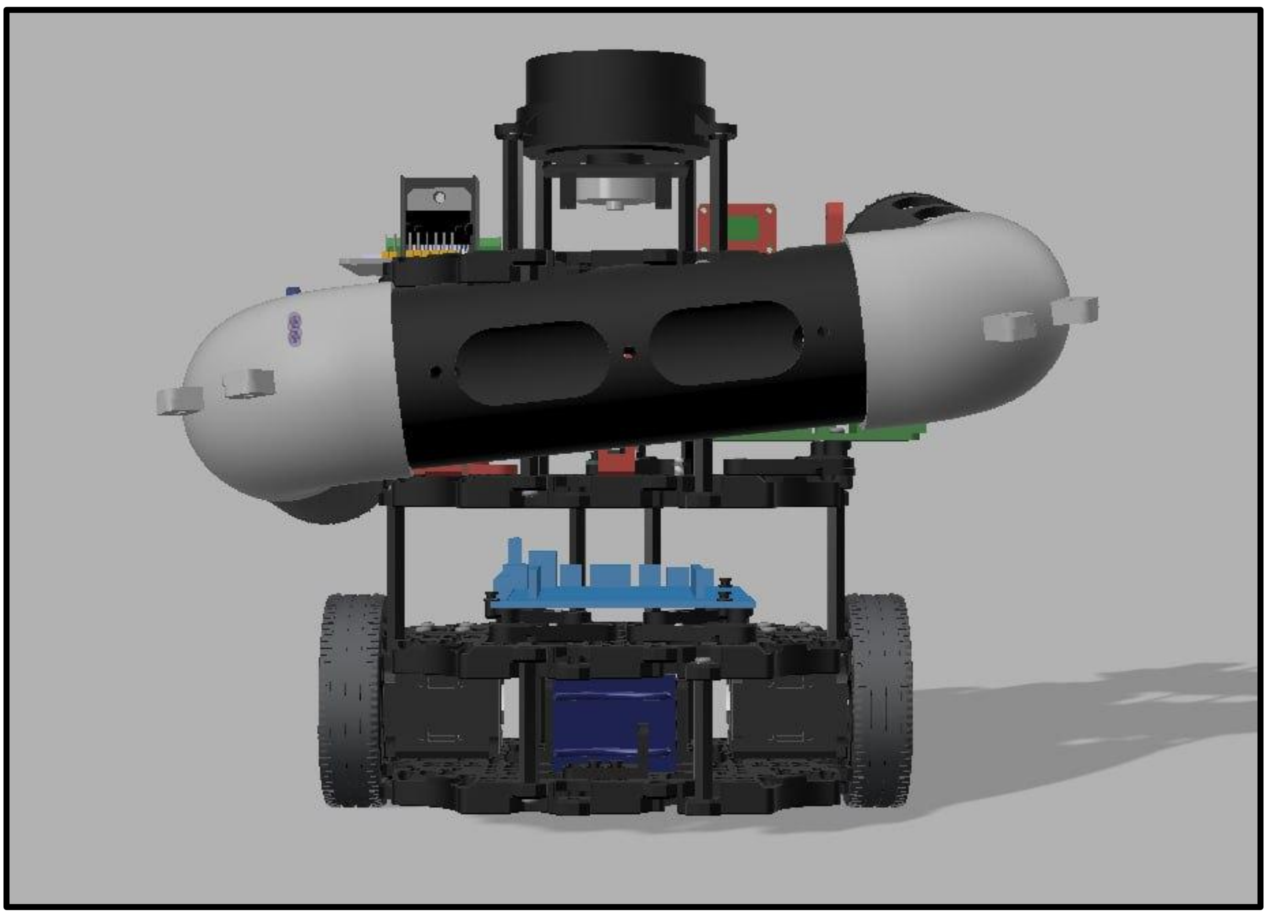
Section 1: General System Description & Spec Sheet:









Rear view of the robot

Running time: 1 hour 18 minutes (on AutoNav)
Charging time: 1 hour
Centre of Gravity: 12.9, 26.2, -90.3 (WRT geometric center of ground plane)

Overall bot based on the Turtlebot 3 overall subsystem:

Electronic Components	Li-Po Battery 11.1V 1800mAh DC-DC Step-down Buck Converter L298N Motor driver
Actuators Used	Sprocket Wheels MG90S Servo Motor RS380 DC Motor Dynamixel Motors
Microcontrollers	Raspberry Pi 4 Open CR 1.0
Sensors used	LDS-02 LiDAR 360 RPI Camera 2
Software	Ubuntu 22.04.5

Bot Purpose: Parcel delivery system to auto-detect target location (both static and dynamic) and deliver payload using automatic navigation through a maze-like course.

Section 2: Technical Guide

1. To set up operational environment:
    - Paste ArUco markers 0 and 1 on the right side of the static and dynamic stations' target holes respectively
    - Paste ArUco marker 2 over the dynamic stations's target hole
    - Paste ArUco marker 3 at the lift
2. Load table tennis balls into the launcher tubes
3. Connect battery to female battery terminals
4. Turn on OpenCR board using the switch

To launch program:

SSH Into RPI Terminal
1) fullbu
Local Terminal
1) full_run

Section 3: Acceptable Defect Log

S. No	Defect Description	Risk Classification Level		
		Low	Medium	HIgh
1	Servo motor is twitching but it is able to consistently hold back the table tennis balls			
2	Flywheel is attached to motor shaft using only friction			
3	Servo Jumper wires connected using M-F jumpers (terminal blocks not used)			
4	DC Motor magnet may attract wires, causing short circuits through battery terminals			

Risk Mitigation Strategy: -

Risk No.	Strategy Imposed
1	Different power sources for servo considered (using buck converter, RPi pinouts, etc.); secured screws to minimise twitching effect
2	Used hot glue gun on shaft head
3	Wires spaced out as much as possible
4	DC motor, and all other metallic parts securely attached

Section 4: Factory Acceptance Test

s/n	Check item	Pass	Remarks
Visual Inspection (Powered Off)			
1	**Flywheel security**: Ensure the flywheel is securely on and the flywheel spins freely by hand with no obstruction		If the fit is loose, use hot glue or masking tape for a more secure fit.
2	**Flywheel motor mount**: Ensure mounting screws are tight and motor does not shift		

3	**Servo motor mount**: Ensure mounting screws are tight and motor does not shift		
4	**Cable security**: Ensure all cables are plugged in with no loose unplugged ones		
Power-on Idle Checks:-			
5	**Power-on LEDs lit:** RPi, OpenCR board, L293D Motor Driver all have their LEDs turned on		If not working, verify that the OpenCR switch is on, the motor driver wires are connected and the RPI is connected to OpenCR
6	**Camera Functions**: SSH into the Turtlebot3 and run camera bringup command. On your local terminal run rqt. Ensure camera feed is coloured and non-static		
7	**Mobility**: Run rteleop and test movement in all Directions		
8	**Overall system power:** Check for the LDS02 spinning to ensure power distribution through all microcontrollers		

Section 5: Maintenance and Part Replacement Log

- Test battery cell voltage before and after sessions, ensure it does not drop under 3.3V per cell
- Ensure flywheel motor is at correct height by launching a ball
- Ensure servo is correctly positioned by launching a ball
- Ensure tubes are properly attached by lifting the robot by the tube and observing stability

Date	Incident	Rectification	Remarks
07/04/2026	The battery could not charge anymore	Battery Replaced	Don’t let the battery get too low
29/03/2026	Motor magnet caused a loose spacer to connect across battery terminals	Battery Replaced	Don’t keep all your eggs in one basket